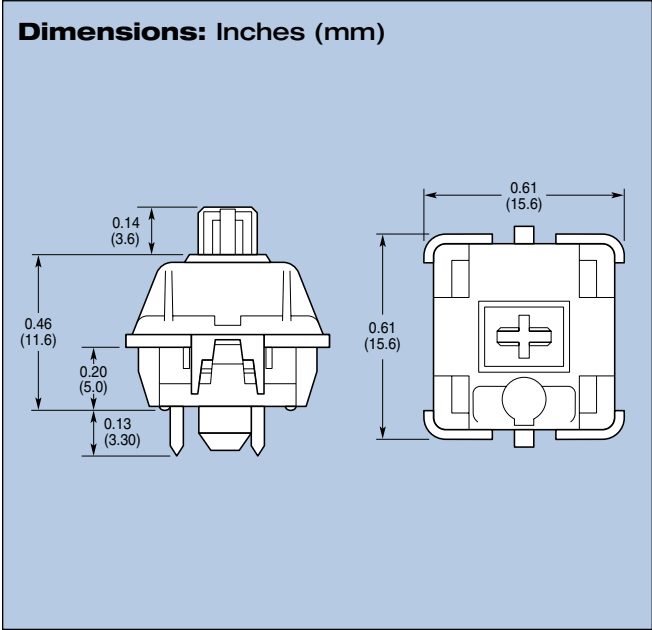
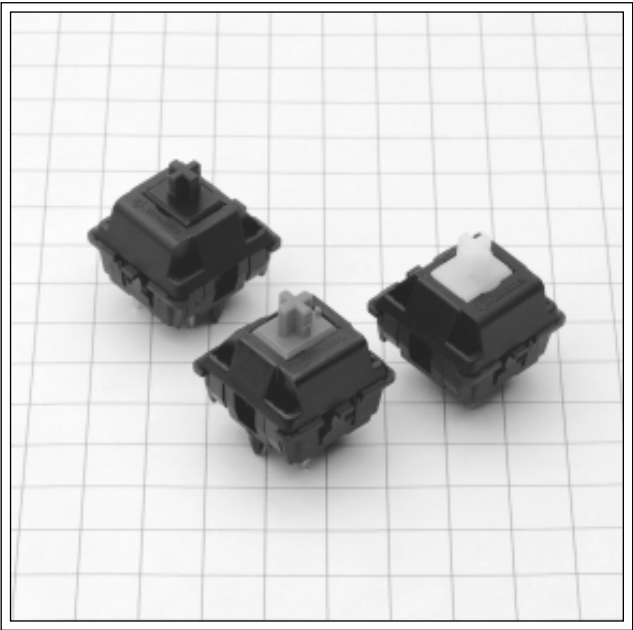


MX Series

Keyswitch

MX Desktop Profile



Specifications

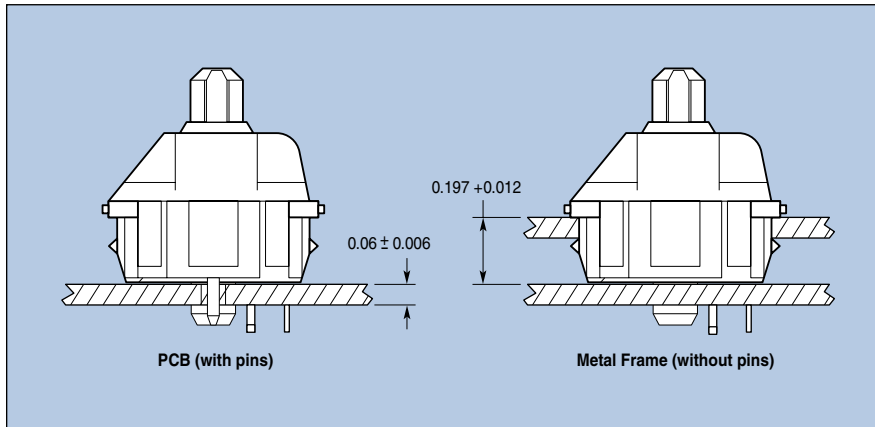
Voltage	12 VAC/DC max. 2 VDC min.
Current	10mA AC/DC max. 10µA DC min.
Insulation Resistance	100 M Ω at 100 VDC
Capacitance	<2pF (at 1kHz)
Bounce Time	$\leq$ 5 msec (at 16 in/sec actuation speed)
Life	(at 5 V, 1mA, no load)
MX Linear	50 x 10 ⁶ cycles
MX Soft Tactile	20 x 10 ⁶ cycles
MX Click Tactile	20 x 10 ⁶ cycles
MX Alternate Action	50 x 10 ⁵ cycles
Initial Contact Resist.	200 m Ω (25 m Ω typical)
Materials	
Plastics	Thermoplastic, UL-rated
Contacts	AuAg 10
Spring	Stainless steel
Flammability Rating	UL94HB
Solderability	Wave solder 5 seconds at 500°F
Environmental	
Operating Temp. Range	-10° to +70°C
Storage Temp. Range	-40° to +70°C
Relative Humidity	5% to 95%, non-condensing

Mechanical Motion

	Linear Feel	Soft Tactile Feel
Operating Force	2.1 oz $\pm$ 0.7 oz	2.0 oz $\pm$ 0.7 oz
Pretravel	0.08" $\pm$ 0.24"	0.08" $\pm$ 0.24"
Total Travel	0.16" -0.015"	0.16" -0.2"
Tactile Force	N/A	2.3 oz $\pm$ 0.7 oz

	Click Tactile Feel	Alternate Action
Operating Force	1.8 oz $\pm$ 0.5 oz	2.1 oz $\pm$ 0.7 oz
Pretravel	0.09" $\pm$ 0.24"	0.05" $\pm$ 0.015"
Total Travel	0.16" -0.02"	0.16" $\pm$ 0.01"
Tactile Force	2.1 oz $\pm$ 0.5 oz	N/A

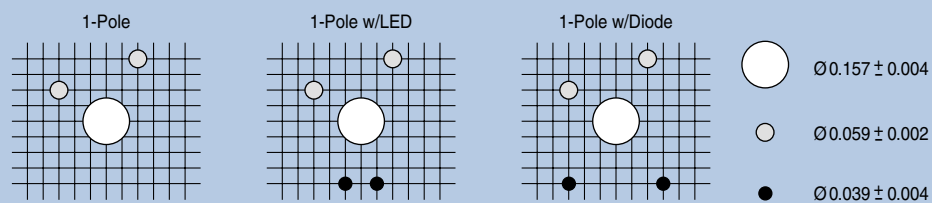
Mounting Options



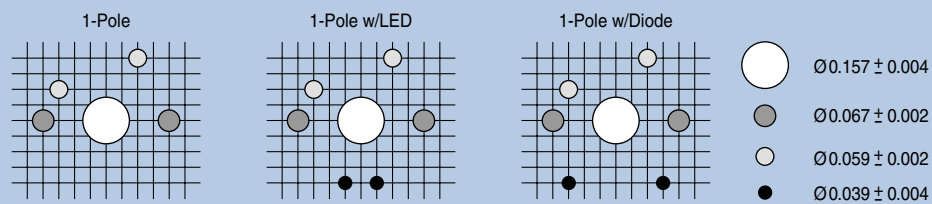
Circuit Board Layouts

Grid line spacing = 0.05 inch

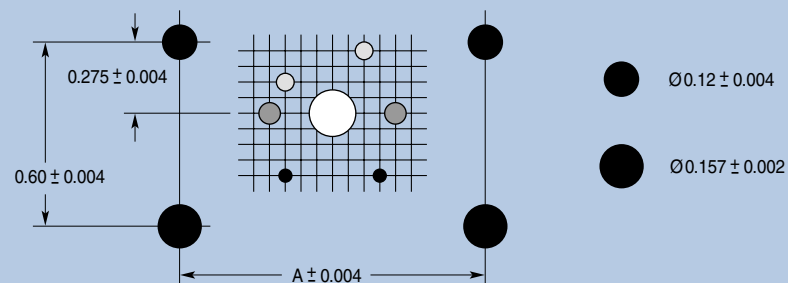
Keypress without fixation pins



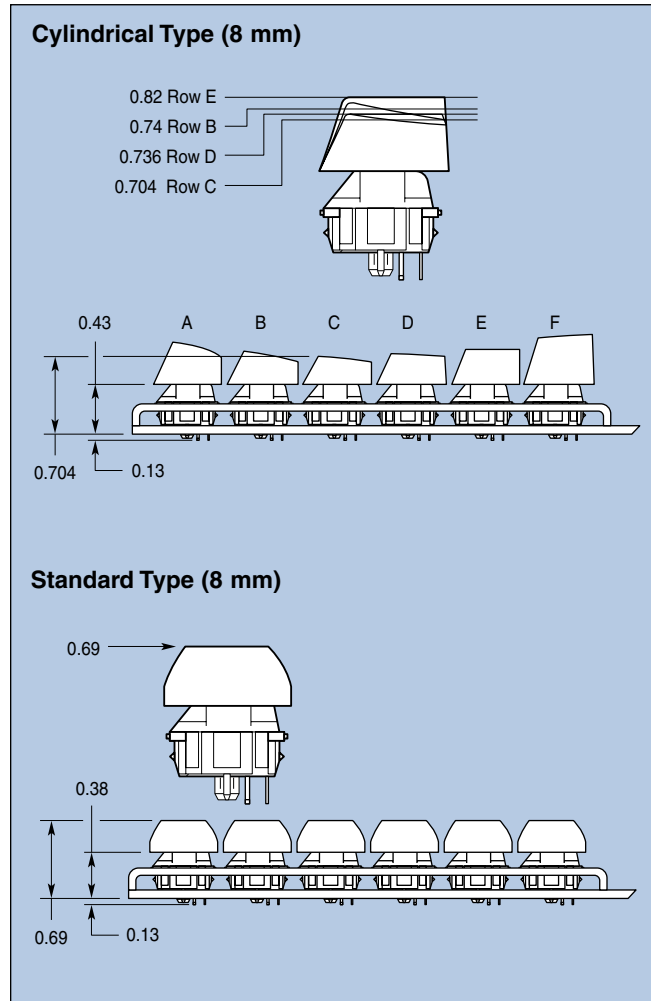
Keypress with fixation pins



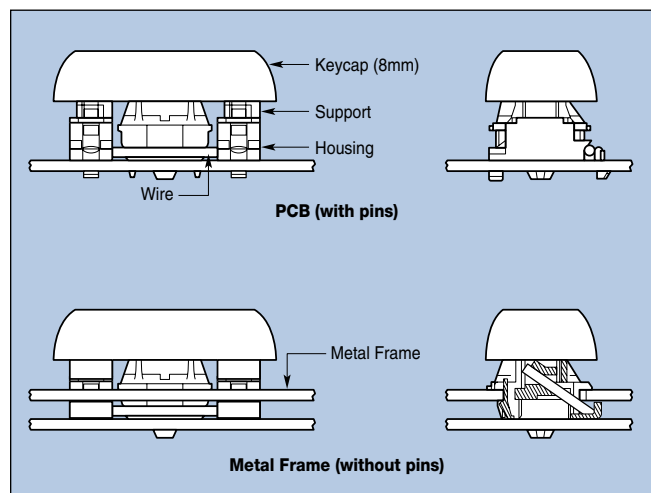
Spacebar without frame



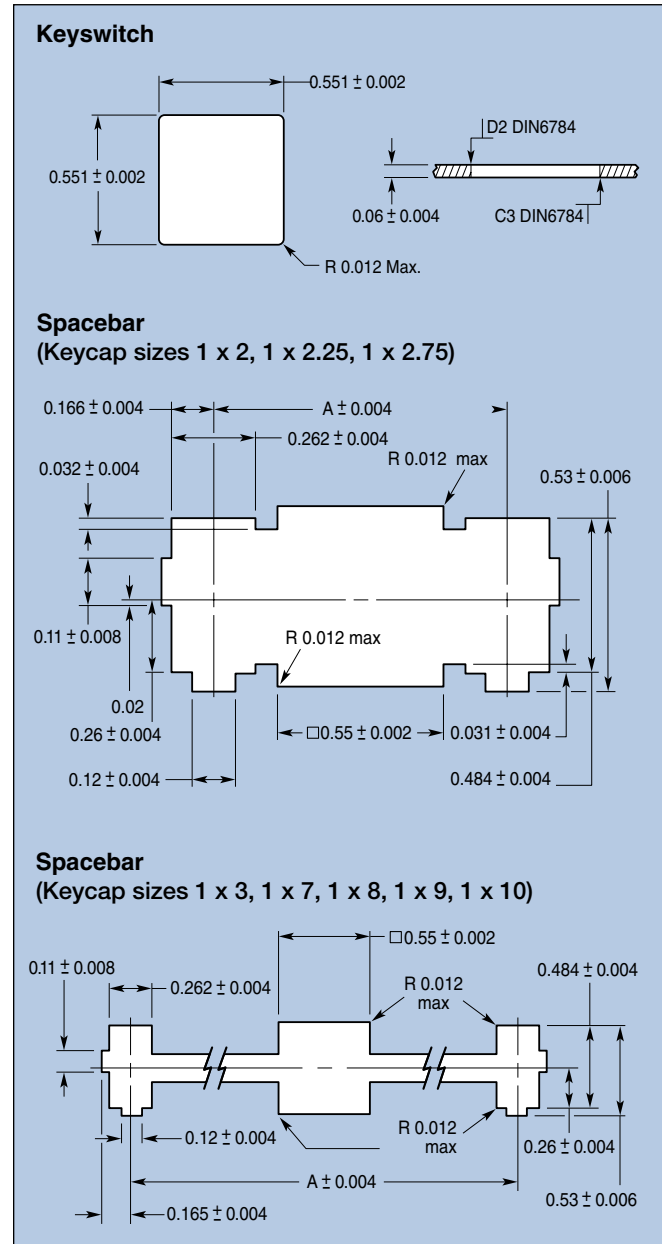
Keycap Dimensions



Spacebar Mounting Options



Metal Frame Cutout Dimensions



Leveling Mechanism Kits

Keycap Size	1 x 2 1 x 2, 2.5 1 x 2, 2.75	1 x 3	1 x 8	1 x 8 1 x 9 1 x 10	1 x 7
Keycap Type	Std & Cylin	Std & Cylin	Std	Cylin	Cylin
A (inches)	0.94	1.5	5.25	5.25	4.5
P/N w/frame	G99-0224	G99-0225	G99-0226	G99-0226	G99-0379
P/N w/o frame	G99-0742	G99-0743	G99-0744	G99-0744	G99-0745

Keyswitch

Ordering Information

Part Number	Description
-------------	-------------

- | | |
|-----------|---|
| MX1A-11DN | Standard force, momentary face plate mount with diode |
| MX1A-11NN | Standard force, linear, face plate mount |
| MX1A-11NW | Standard force, linear, PCB mount |
| MX1A-21NW | Higher force, linear, PCB mount |
| MX1A-31NN | Alternate action, face plate mount |
| MX1A-31NW | Alternate action, PCB mount |
| MX1A-C1NN | Standard force, soft tactile, face plate mount |
| MX1A-C1NW | Standard force, soft tactile, PCB mount |
| MX1A-E1NN | Standard force, click tactile, face plate mount |
| MX1A-E1NW | Standard force, click tactile, PCB mount |

MX1A-C1NN	Standard force, soft tactile, face plate mount
MX1A-C1NW	Standard force, soft tactile, PCB mount
MX1A-E1NN	Standard force, click tactile, face plate mount
MX1A-E1NW	Standard force, click tactile, PCB mount

- | | |
|-----------|---|
| MX1A-C1NN | Standard force, soft tactile, face plate mount |
| MX1A-C1NW | Standard force, soft tactile, PCB mount |
| MX1A-E1NN | Standard force, click tactile, face plate mount |
| MX1A-E1NW | Standard force, click tactile, PCB mount |

- | | |
|-----------|---|
| MX1A-C1NN | Standard force, soft tactile, face plate mount |
| MX1A-C1NW | Standard force, soft tactile, PCB mount |
| MX1A-E1NN | Standard force, click tactile, face plate mount |
| MX1A-E1NW | Standard force, click tactile, PCB mount |

MX1A-C1NN	Standard force, soft tactile, face plate mount
MX1A-C1NW	Standard force, soft tactile, PCB mount
MX1A-E1NN	Standard force, click tactile, face plate mount
MX1A-E1NW	Standard force, click tactile, PCB mount

- | | |
|-----------|---|
| MX1A-C1NN | Standard force, soft tactile, face plate mount |
| MX1A-C1NW | Standard force, soft tactile, PCB mount |
| MX1A-E1NN | Standard force, click tactile, face plate mount |
| MX1A-E1NW | Standard force, click tactile, PCB mount |

MX1A-C1NN	Standard force, soft tactile, face plate mount
MX1A-C1NW	Standard force, soft tactile, PCB mount
MX1A-E1NN	Standard force, click tactile, face plate mount
MX1A-E1NW	Standard force, click tactile, PCB mount

- | | |
|-----------|---|
| MX1A-C1NN | Standard force, soft tactile, face plate mount |
| MX1A-C1NW | Standard force, soft tactile, PCB mount |
| MX1A-E1NN | Standard force, click tactile, face plate mount |
| MX1A-E1NW | Standard force, click tactile, PCB mount |

MX1A-C1NN	Standard force, soft tactile, face plate mount
MX1A-C1NW	Standard force, soft tactile, PCB mount
MX1A-E1NN	Standard force, click tactile, face plate mount
MX1A-E1NW	Standard force, click tactile, PCB mount

